Full-Support Regularization with Uniform Entropy Control

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Abstract

Every process on a finite alphabet admits an arbitrarily small total-variation perturbation whose finite conditional probabilities are all strictly positive, while preserving its total atomic mass and the probabilities of all tail events. The perturbation increases every finite-prefix entropy by less than an arbitrarily prescribed constant, uniformly over the horizon. It is implemented by an independent countably supported seed of arbitrarily small Shannon entropy and changes only finitely many coordinates almost surely. Consequently, atomic-capacity and entropy-growth constructions for one-action processes persist under strict positivity of all finite transitions. Historical priority and Lean formalization are not asserted.

1 Setting

Fix an integer $d \geq 2$ and identify the output alphabet with the finite cyclic group $\mathbb{Z}_d$. Let

$$\Omega = \mathbb{Z}_d^{\mathbb{N}}$$

carry its product σ-algebra. For $x \in \Omega$, write $x_1^n = (x_1, \ldots, x_n)$.

All entropies and logarithms are in base two. For probability measures on Ω, use

$$d_{\mathrm{TV}}(\mu, \nu) = \sup_A |\mu(A) - \nu(A)|.$$

Write

$$a(\mu) = \sum_{x \in \Omega} \mu(\{x\})$$

for the total atomic mass; only countably many summands can be positive.

The tail σ-algebra is

$$\mathcal{T} = \bigcap_{N \geq 1} \sigma(x_N, x_{N+1}, \ldots).$$

A measure has *full finite-prefix support* if every cylinder $\{x : x_1^n = w\}$ has positive probability.

2 Regularization theorem

Theorem 1 (Entropy-controlled full-support regularization)**.** *Let X be any Ω-valued random variable and let $\delta > 0$. There exists a coupling of X with an Ω-valued random variable Y satisfying all of the following:*

1. *Every finite word has positive probability under Y:*

$$\mathbb{P}(Y_1^n = w) > 0 \qquad (n \geq 1,\ w \in \mathbb{Z}_d^n).$$

Consequently, every next-symbol conditional probability after every finite history is strictly positive.

2. *For every $n \geq 1$,*

$$\boxed{0 \leq H(Y_1^n) - H(X_1^n) < \delta.}$$

3. *The total atomic mass is exactly preserved:*

$$\boxed{a(\mathcal{L}(Y)) = a(\mathcal{L}(X)).}$$

4. *Almost surely, X and Y eventually agree:*

$$\mathbb{P}\big(\exists N\ \forall n \geq N :\ X_n = Y_n\big) = 1.$$

In particular,

$$\mathbb{P}(X \in A) = \mathbb{P}(Y \in A) \qquad (A \in \mathcal{T}).$$

5. *Their entire infinite-path laws satisfy*

$$d_{\mathrm{TV}}(\mathcal{L}(X), \mathcal{L}(Y)) < \delta.$$

Moreover, $Y = X + E$ coordinatewise modulo d, where E is independent of X, countably supported, and satisfies $H(E) < \delta$.

Proof. Let

$$\mathcal{E} = \left\{ e \in \mathbb{Z}_d^{\mathbb{N}} : 0 < |\{n : e_n \neq 0\}| < \infty \right\}.$$

This is a countably infinite set. Enumerate it without repetition as $e^{(1)}, e^{(2)}, \ldots$.

Choose $p \in (0, 1)$ sufficiently small that

$$p < \delta, \qquad h_2(p) + 2p < \delta, \tag{1}$$

where h_2 is binary entropy. Independently of X, choose E with distribution

$$\mathbb{P}(E = 0) = 1 - p, \qquad \mathbb{P}(E = e^{(k)}) = p\, 2^{-k}, \quad k \geq 1.$$

Since $\sum_{k \geq 1} k 2^{-k} = 2$,

$$H(E) = h_2(p) + 2p < \delta. \tag{2}$$

Define

$$Y = X + E \pmod{d}.$$

Full finite-prefix support. Fix $n \geq 1$ and $y \in \mathbb{Z}_d^n$. Some $x \in \mathbb{Z}_d^n$ has $\mathbb{P}(X_1^n = x) > 0$. Let e have prefix $y - x$ and zero entries after time n. Whether $e = 0$ or $e \in \mathcal{E}$, its probability is positive. Independence gives

$$\mathbb{P}(Y_1^n = y) \geq \mathbb{P}(X_1^n = x)\mathbb{P}(E = e) > 0.$$

For every history $w \in \mathbb{Z}_d^n$ and symbol $b \in \mathbb{Z}_d$, both the numerator and denominator in

$$\mathbb{P}(Y_{n+1} = b \mid Y_1^n = w) = \frac{\mathbb{P}(Y_1^{n+1} = wb)}{\mathbb{P}(Y_1^n = w)}$$

are therefore positive.

Uniform entropy control. For each fixed e, translation by e_1^n is a permutation of $\mathbb{Z}_d^n$. Since E is independent of X,

$$H(Y_1^n \mid E) = H(X_1^n).$$

Conditioning cannot increase entropy, so

$$H(Y_1^n) \geq H(X_1^n).$$

Conversely, Y_1^n is a deterministic function of (X_1^n, E), and hence

$$H(Y_1^n) \leq H(X_1^n, E) = H(X_1^n) + H(E).$$

Equation (2) proves the claimed bound simultaneously for every n.

Preservation of atomic mass. Write the path law as

$$\mu = \mu_{\mathrm{at}} + \mu_{\mathrm{na}},$$

where μ_{at} is supported on the countable set of point atoms and μ_{na} has no point atoms. Let $a = \mu_{\mathrm{at}}(\Omega)$.

For a finite-support sequence e, let $T_e(x) = x + e$. Each T_e is a measurable bijection. Writing $q_e = \mathbb{P}(E = e)$,

$$\nu := \mathcal{L}(Y) = \sum_e q_e (T_e)_* \mu_{\mathrm{at}} + \sum_e q_e (T_e)_* \mu_{\mathrm{na}}.$$

The first sum is supported on a countable union of countable sets and has total mass a. For every $y \in \Omega$, the second sum assigns singleton mass

$$\sum_e q_e \, \mu_{\mathrm{na}}(\{T_e^{-1} y\}) = 0.$$

Thus this is precisely an atomic–nonatomic decomposition, and $a(\nu) = a(\mu)$.

Tail events. Every realization of E has finite support. Therefore X and Y eventually agree almost surely.

Every $A \in \mathcal{T}$ is invariant under finite changes of coordinates: if two sequences agree from time N onward, membership in A, being measurable with respect to those coordinates, is the same for both. Consequently,

$$\mathbf{1}_A(X) = \mathbf{1}_A(Y) \quad \text{almost surely.}$$

Total variation. On $\{E = 0\}$, the entire trajectories agree. The coupling inequality and (1) give

$$d_{\mathrm{TV}}(\mu, \nu) \leq \mathbb{P}(X \neq Y) = \mathbb{P}(E \neq 0) = p < \delta.$$

□

3 Identification with one-action atomic capacity

For a probability law P on a finite set, let $A_K(P)$ be the mass of its K largest atoms, using all atoms if fewer than K exist.

For a path law μ, let μ_n denote its length-n prefix law and define

$$\mathrm{atomQ}(\mu) = \sup_{K \geq 1} \inf_{n \geq 1} A_K(\mu_n).$$

Lemma 1. *For every probability measure μ on Ω,*

$$\mathrm{atomQ}(\mu) = a(\mu).$$

Proof. Given $\eta > 0$, finitely many point atoms of μ have total mass at least $a(\mu) - \eta$. If their number is K, their prefixes occupy at most K cylinders at every horizon. Hence

$$A_K(\mu_n) \geq a(\mu) - \eta \qquad \text{for every } n,$$

which proves $\mathrm{atomQ}(\mu) \geq a(\mu)$.

For the reverse inequality, write $\mu = \mu_{\mathrm{at}} + \mu_{\mathrm{na}}$ and set

$$b_n = \max_{w \in \mathbb{Z}_d^n} \mu_{\mathrm{na}}([w]).$$

We claim $b_n \to 0$. Otherwise, for some $\eta > 0$, there are cylinders of arbitrarily large depth with μ_{na}-mass at least η. The collection of such words is a prefix-closed finitely branching tree with unbounded depth. An infinite branch gives nested cylinders $[w_n]$ with mass at least η. Continuity from above then gives a point atom of μ_{na} of mass at least η, a contradiction.

Any union of K length-n cylinders has atomic-component mass at most $a(\mu)$ and nonatomic-component mass at most Kb_n. Therefore

$$A_K(\mu_n) \leq a(\mu) + Kb_n.$$

For each fixed K, taking $n \to \infty$ gives $\inf_n A_K(\mu_n) \leq a(\mu)$. Taking the supremum over K proves the result. □

For a one-action process, the optimal exact finite-horizon seed entropy is

$$C_n = H(X_1^n).$$

Indeed, every exact seed maps deterministically to the transcript, and the transcript itself is an admissible seed.

Corollary 1 (Preservation of capacity and the cost curve)**.** *Every one-action process on d symbols has, for every $\delta > 0$, a strictly positive one-action regularization on the same alphabet satisfying*

$$\boxed{\widetilde{\mathrm{atomQ}} = \mathrm{atomQ}, \qquad 0 \leq \widetilde{C}_n - C_n < \delta \quad \textit{for every } n.}$$

Proof. Apply Theorem 1 and Lemma 1. □

4 Consequences for entropy-growth constructions

Corollary 2 (Transfer of diverging asymptotics)**.** *Suppose a one-action process satisfies $C_n \sim f(n)$, where $f(n) \to \infty$. Its regularization satisfies*

$$\widetilde{C}_n \sim f(n)$$

and has exactly the same atomic capacity. More generally, every asymptotic expansion of C_n up to an $O(1)$ remainder is preserved.

Proof. The entropy difference is uniformly bounded by δ. □

The preceding realization theorem for concave sublinear scales states that, whenever

$$f(0) = 0, \qquad 1 \geq f(n) - f(n-1) \downarrow 0, \qquad f(n) \to \infty,$$

every $a \in [0,1]$ occurs in a one-action process with at most three symbols satisfying

$$\mathrm{atomQ} = a, \qquad C_n \sim f(n).$$

Regularization therefore gives the following strengthening.

Corollary 3 (Strictly positive sublinear realizations)**.** *For every such f and every $a \in [0,1]$, there exists a one-action process on three symbols with*

$$\mathrm{atomQ} = a, \qquad C_n \sim f(n),$$

whose conditional probability of every symbol after every finite history is strictly positive.

Likewise, the one-action constructions realizing the capacity–rate region

$$0 \leq R \leq 1 - a, \qquad R = \lim_{n\to\infty} \frac{C_n}{n \log d},$$

remain valid under strict positivity: regularization preserves the alphabet, the capacity, and the limiting normalized rate.

5 Full support with arbitrarily little seed entropy

Corollary 4. *For every $d \geq 2$ and every $\delta > 0$, there exists a process on $\mathbb{Z}_d$ such that:*

1. *every finite word has positive probability;*
2. *every next-symbol conditional distribution has full support;*
3. *one countably supported seed of entropy less than δ realizes the entire infinite process exactly;*
4. *the process is eventually zero almost surely.*

Proof. Apply the construction to the deterministic zero process $X_n = 0$. Then $Y = E$, and E itself is the required seed. □

Thus strict local uncertainty at every finite history does not, by itself, force an unbounded total entropy requirement.

This does not extend to a uniform positive lower bound on transition probabilities. If, for some fixed $0 < \eta \leq 1/d$,

$$\mathbb{P}(X_{n+1} = b \mid X_1^n = w) \geq \eta$$

for every symbol, time, and history of positive probability, then each conditional entropy is at least

$$h_{\min}(d,\eta) = -\big(1-(d-1)\eta\big)\log\big(1-(d-1)\eta\big) - (d-1)\eta\log\eta > 0.$$

The chain rule yields

$$H(X_1^n) \geq n\, h_{\min}(d,\eta),$$

when the same bound includes the initial distribution. Moreover, every cylinder has mass at most $\big(1-(d-1)\eta\big)^n$, so the infinite-path law is nonatomic.

6 Scope and verification status

The regularization theorem applies to arbitrary finite-alphabet processes, with no stationarity or finite-entropy assumption on the original infinite source. Its proof uses a countable mixture of invertible finite-coordinate translations and standard entropy identities.

The causal-cost conclusions in this note concern one-action processes. No claim is made that an arbitrary multi-action controlled tree admits the same regularization while preserving its optimal common causal seed entropy within δ.

The result establishes that the previously constructed one-action capacity and growth phenomena do not depend on zero-probability branches. It does not identify strict positivity with uniform positivity, nor does it preserve stationarity.

These are mathematical proofs, not Lean-kernel certificates. Historical priority of the combined regularization statement has not been established.

References

[1] J. Mikulík, *One Seed for Every Adaptive Policy*. Mathematical notes and Lean development. `https://github.com/jendamikulik/finite-Shannon-theorem`.

[2] J. Mikulík, *Atomic Capacity and the Entropy Rate of Causal Simulation: A Sharp Feasible-Region Theorem for Bounded Output Alphabets*. Research note, 20 October 2026.

[3] J. Mikulík, *Atomic Capacity Does Not Determine Sublinear Causal Entropy Growth*. Research note, 20 October 2026.